



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



Nicolaus Copernicus
Astronomical Center
of the Polish Academy of Sciences

Radiation Physics for SMBH Accretion Disks in PLUTO

CAMK PAN SUMMER STUDENT PROGRAM

Fritz Ali Agildere

September 11, 2026



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Completed Work

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- Outer Boundary: Extrapolation with Van Leer Limited Slope, Coronal Backflow Guard and Injected Kluźniak-Kita Analytic Disk Solution



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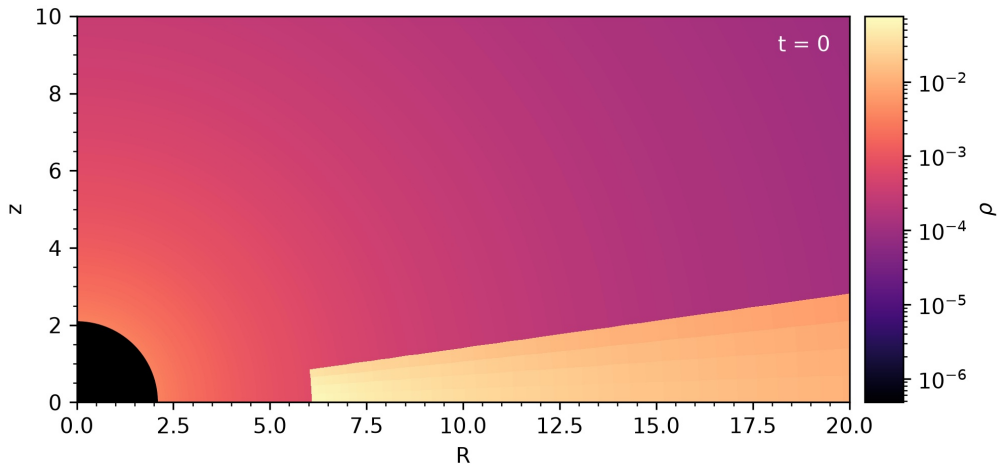
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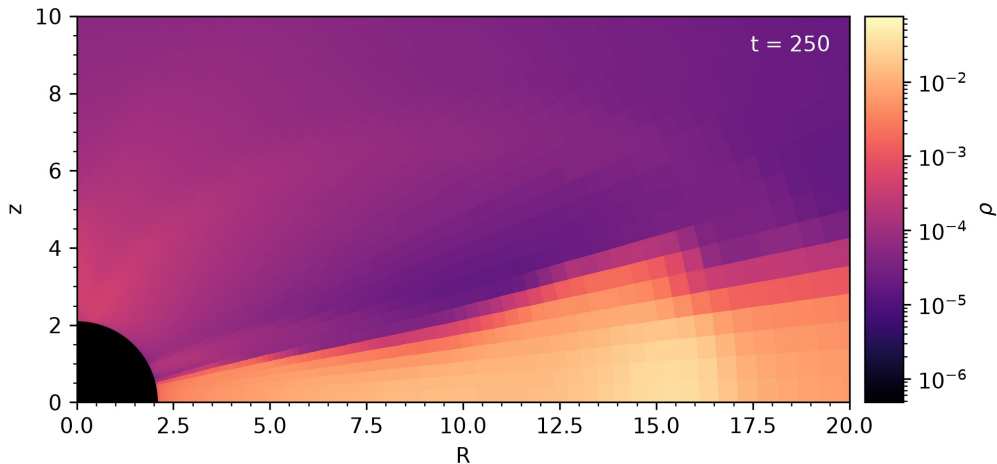
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- Parallelize with MPI Reduction Across Entire Domain and Feeding Back to All Ranks

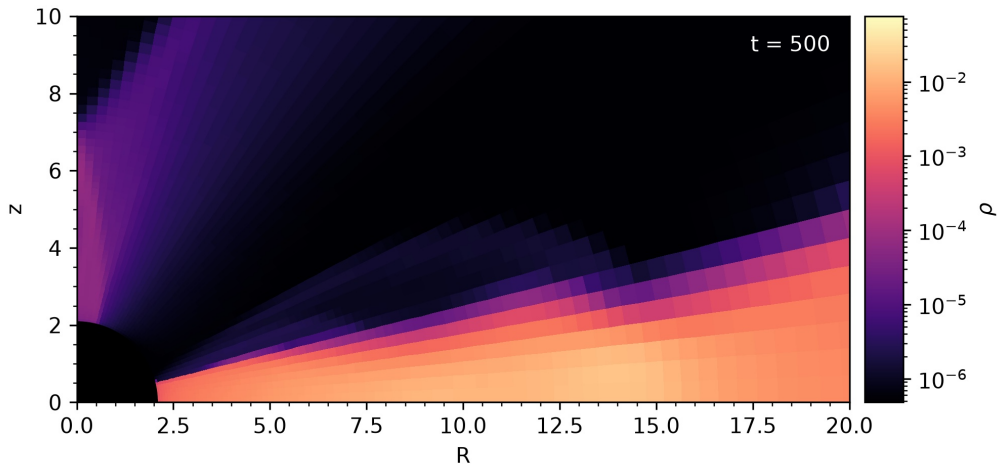
Tracer Physics – Density



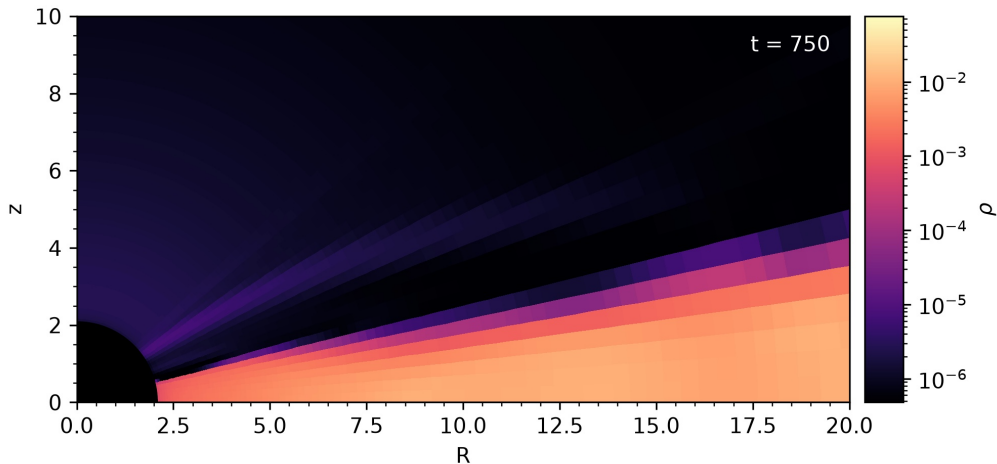
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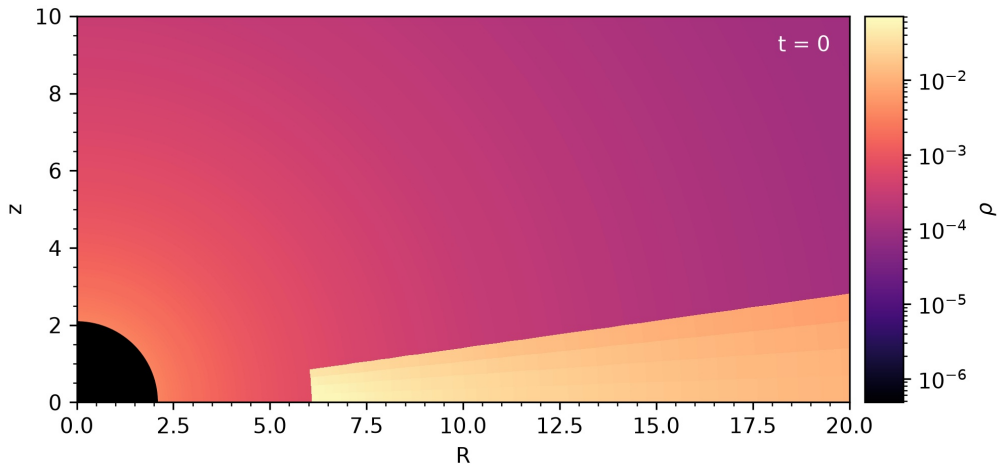
Tracer Physics – Density



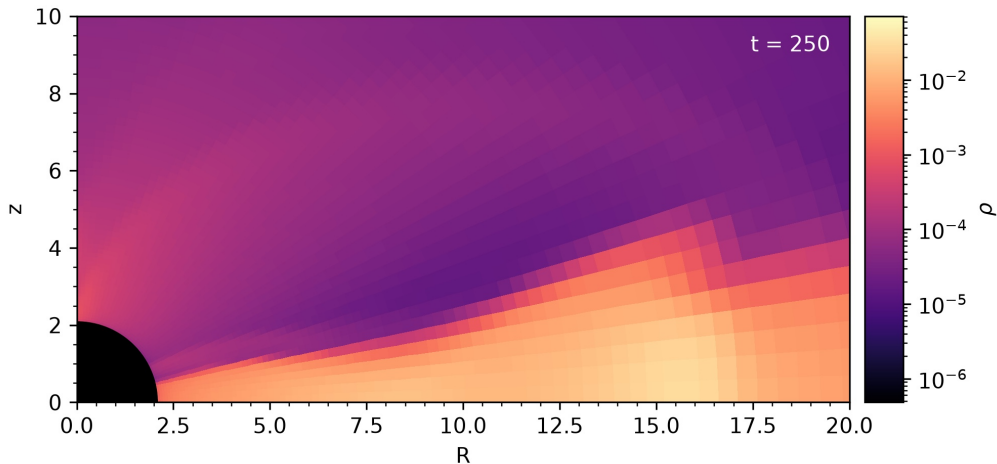
Tracer Physics – Density



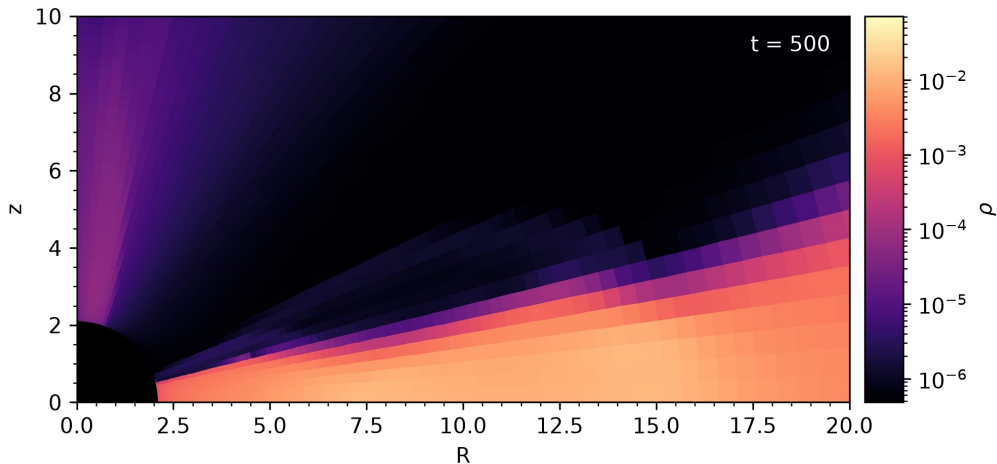
Recovered Physics – Density



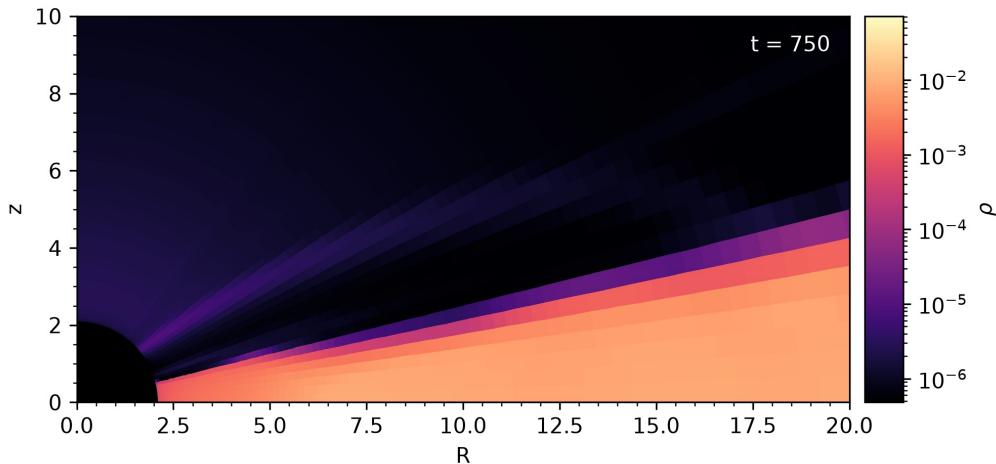
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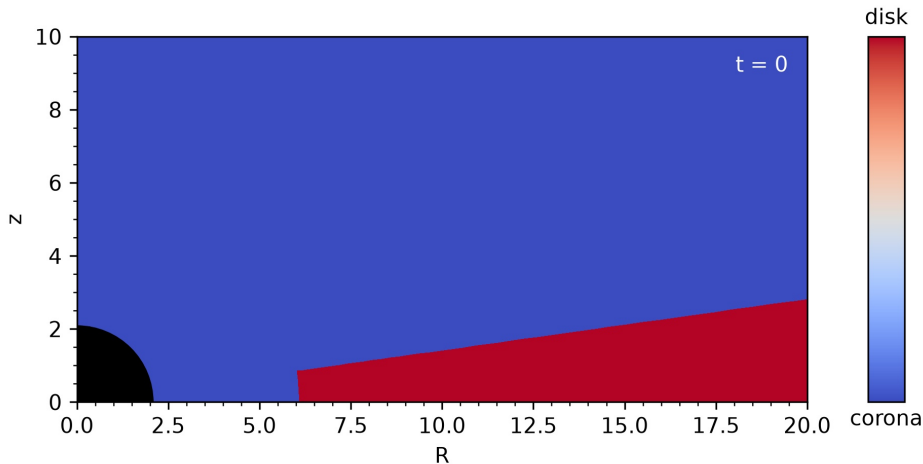
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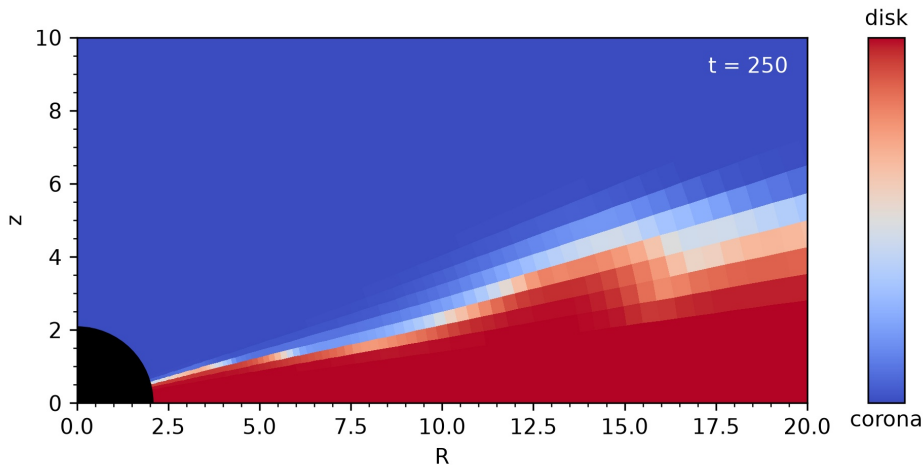
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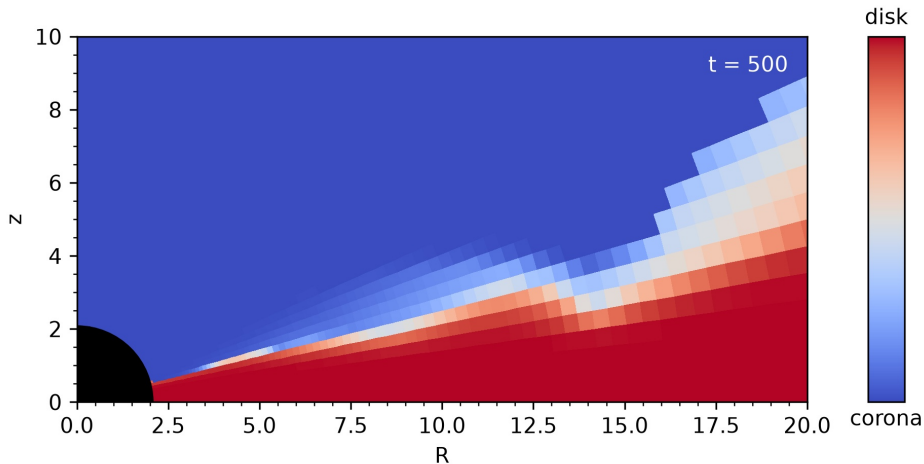
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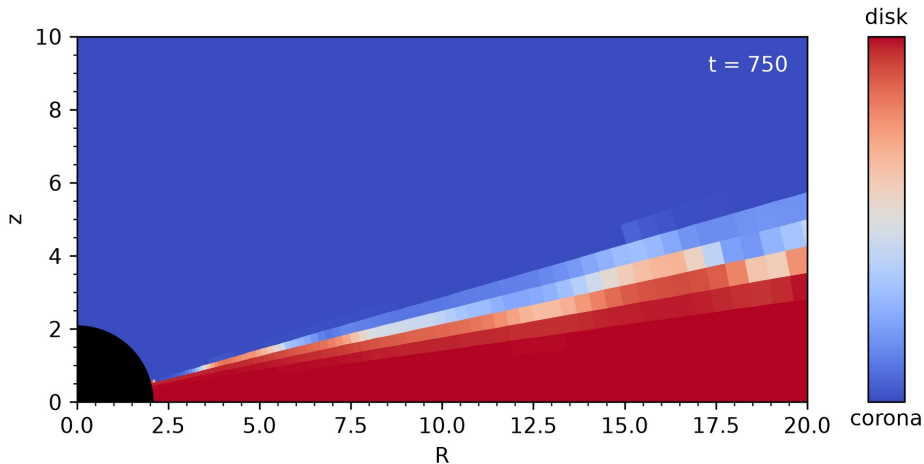
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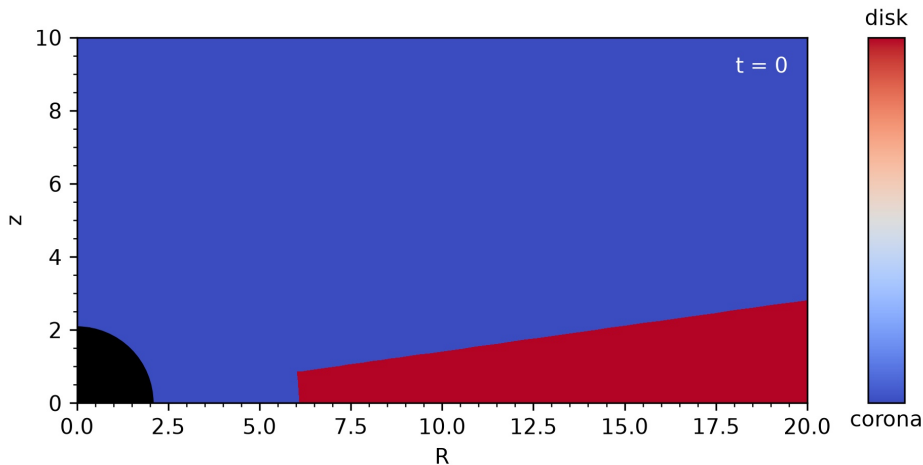


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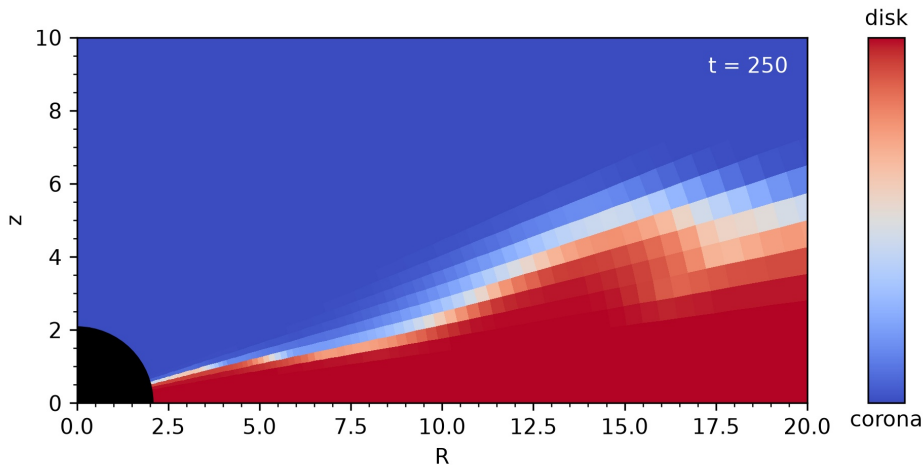


Graphical Comparison

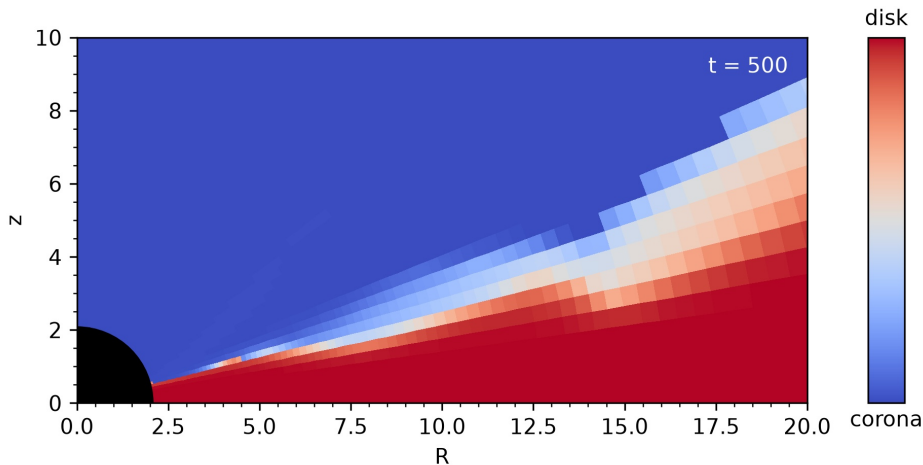
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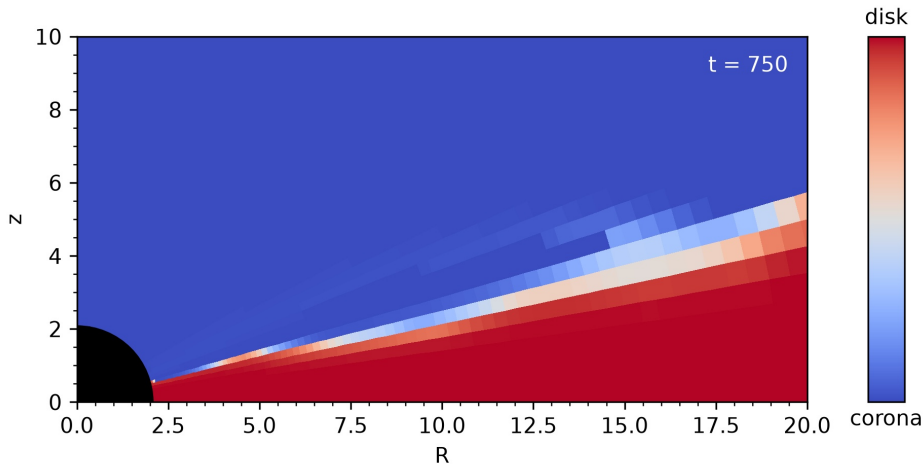


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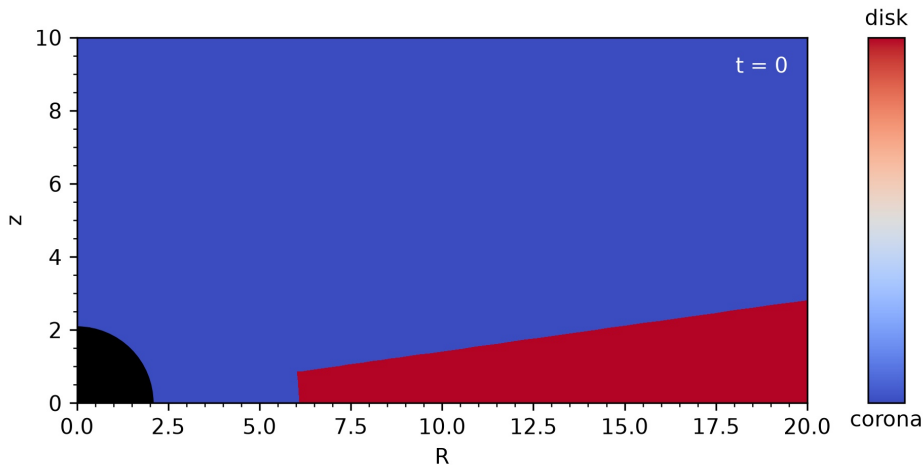


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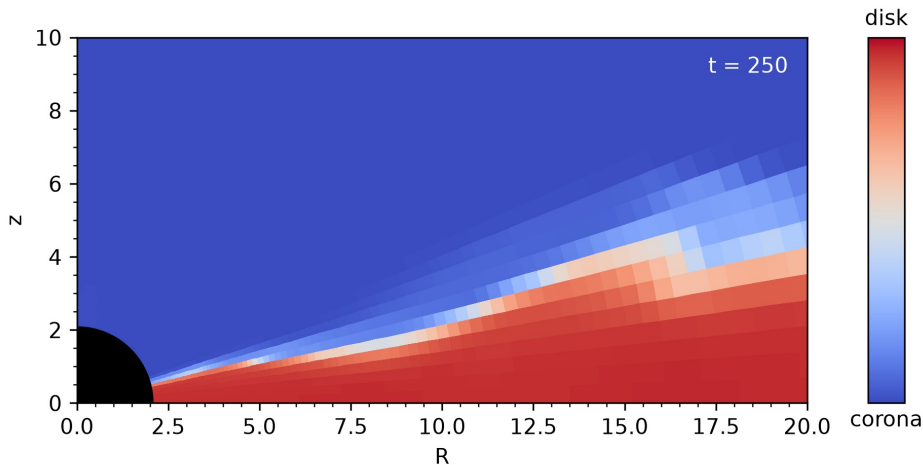
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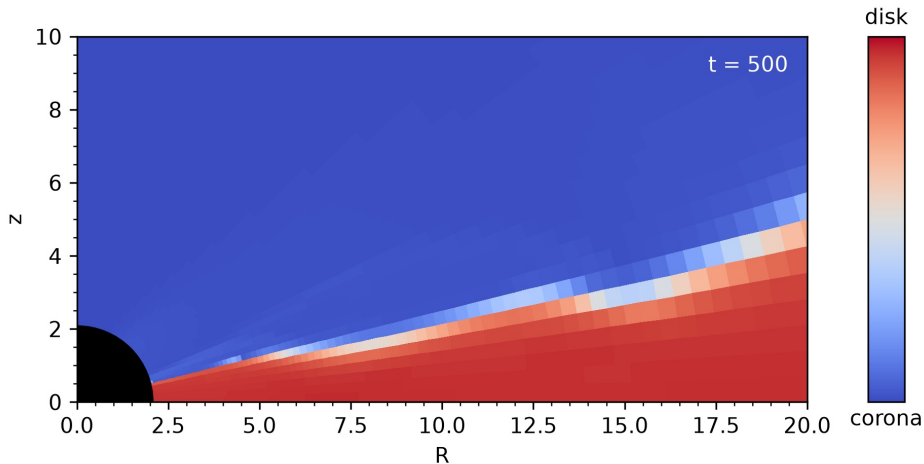
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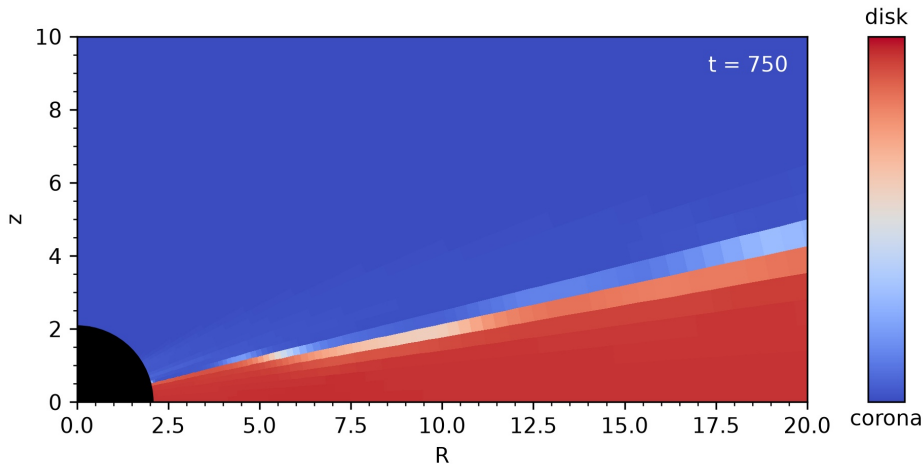
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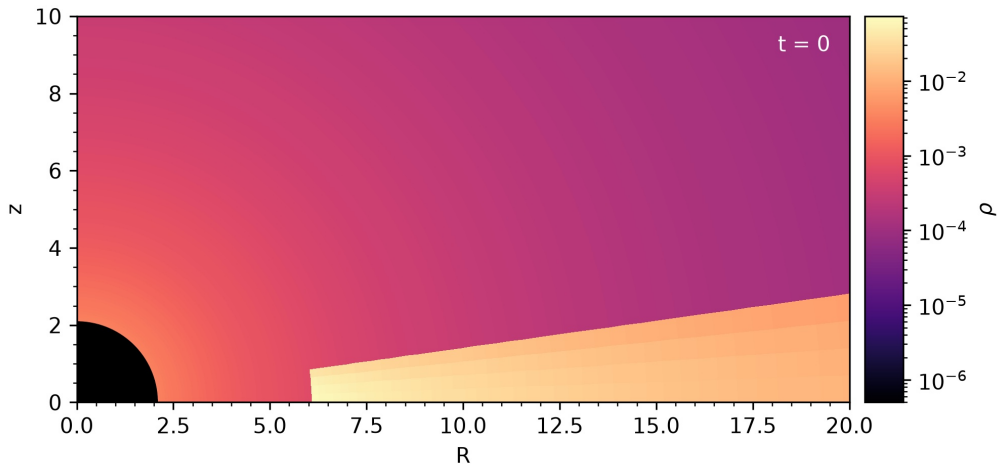
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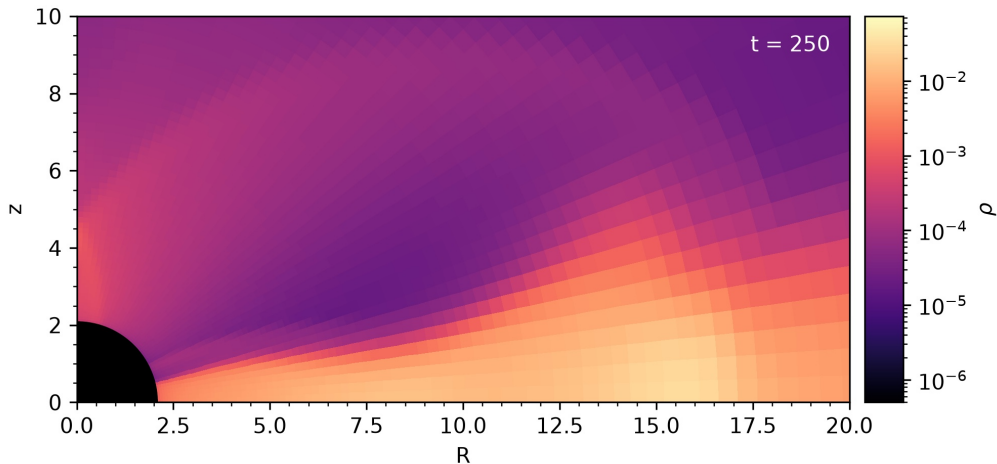
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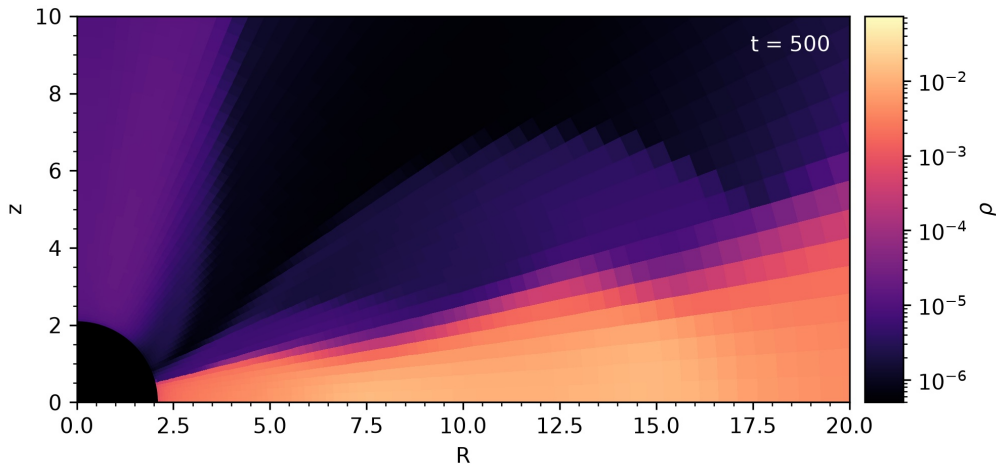
Radiation Enabled – Density



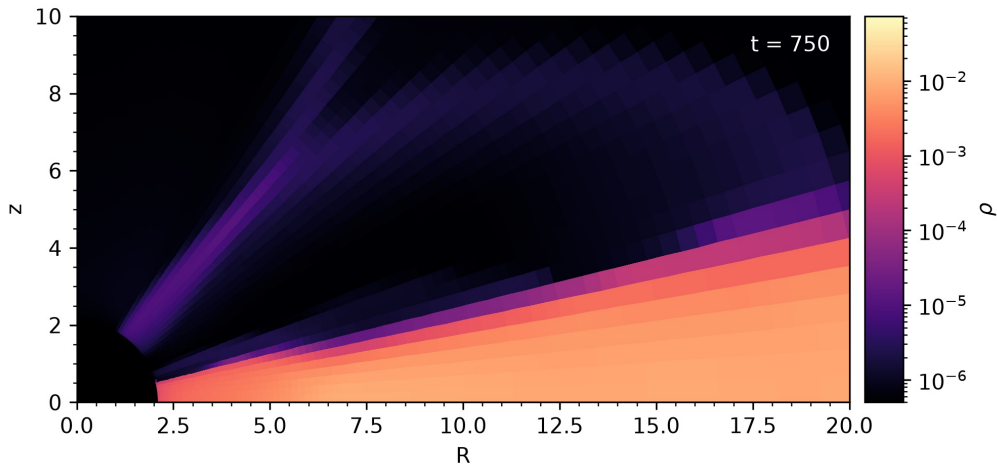
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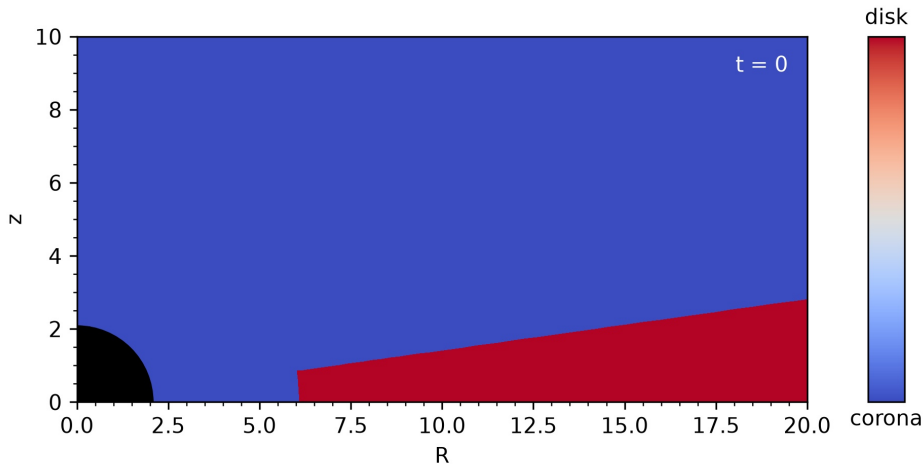
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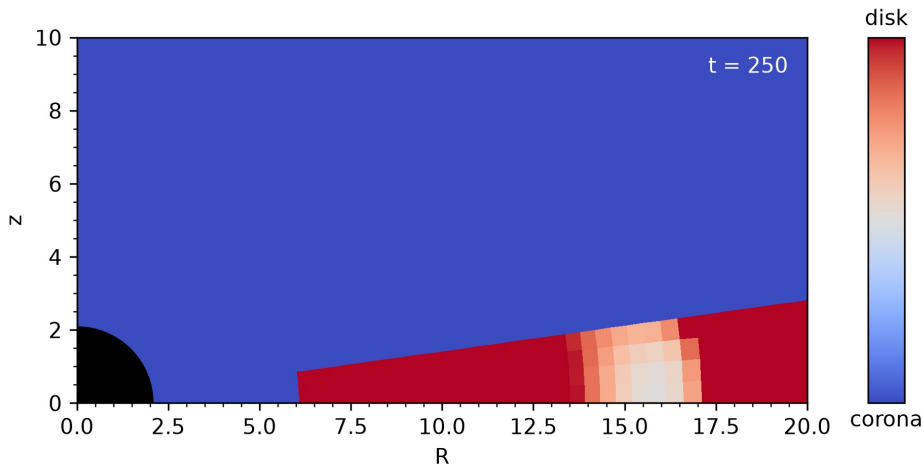
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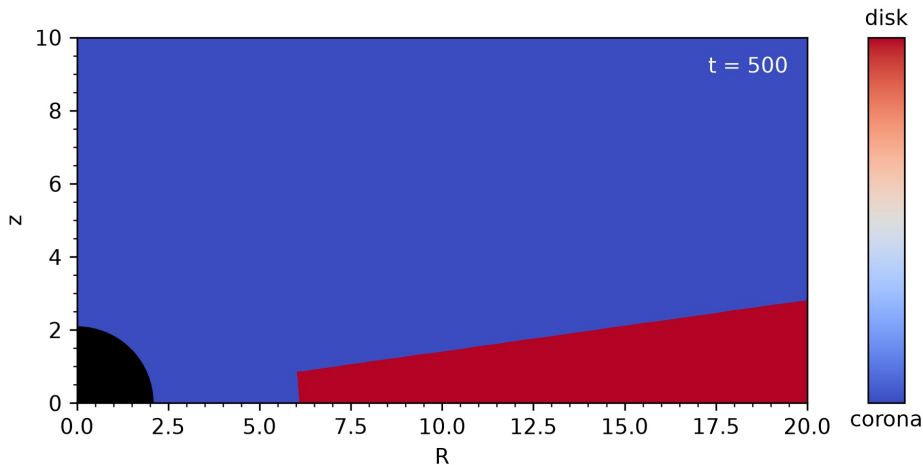
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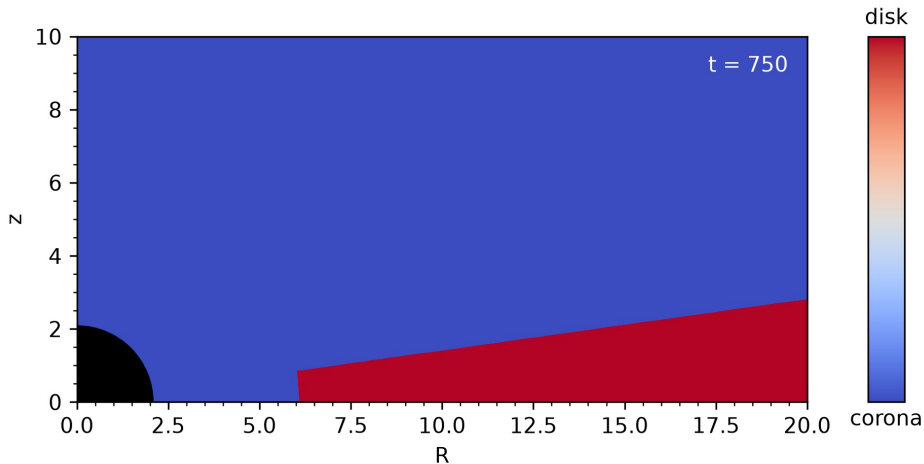
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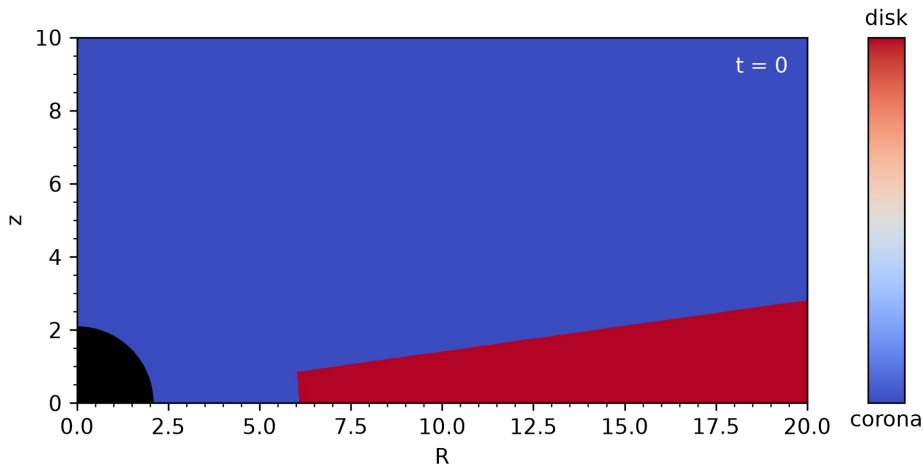
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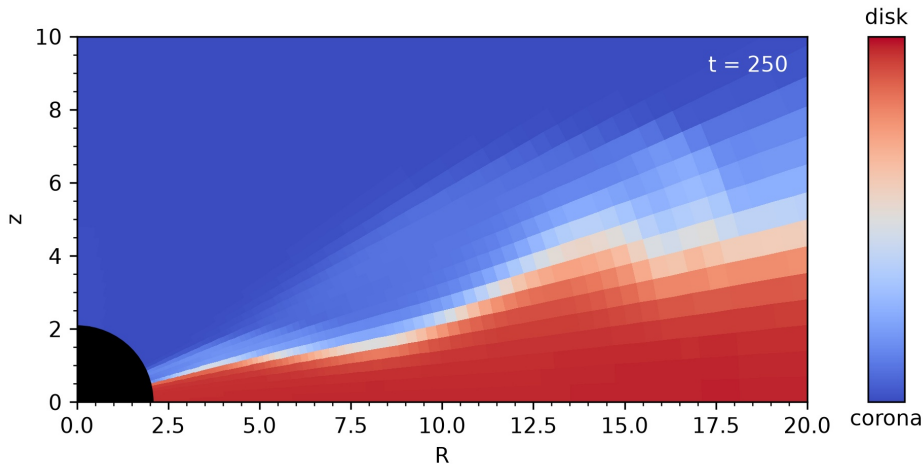
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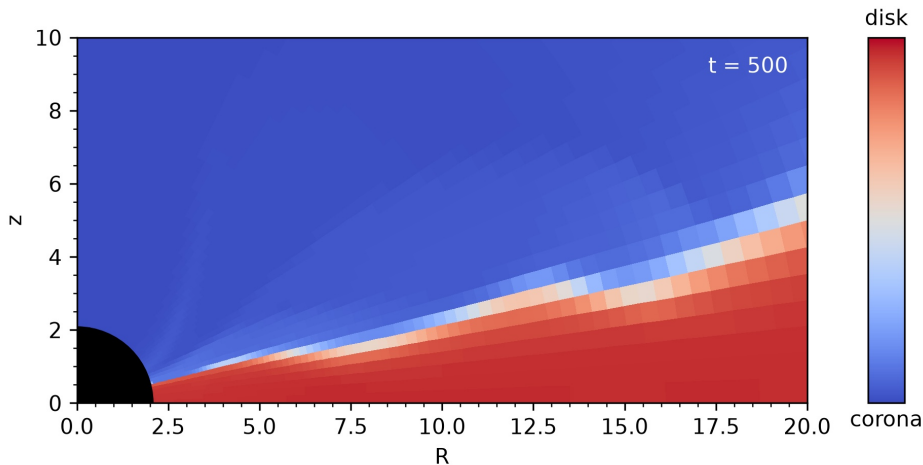
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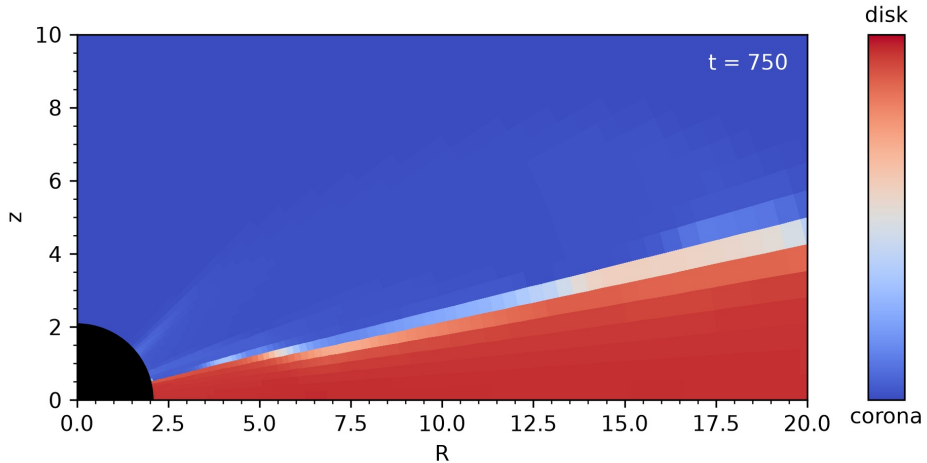


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- Enable Ideal or Resistive MHD Solver and Test Magnetic Field Interactions
- Extend 2.5D to 3D Simulations and Apply Radiation Setup in Different Scenarios

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJ) 2007
- M. Čemeljić, “Atlas” of Numerical Solutions for Star-Disk Magnetospheric Interaction (A&A) 2019
- M. Čemeljić, W. Kluźniak, S. Kundu, *Magnetic Precession as a Model for Quasiperiodic Oscillations in Pulsating Ultra Luminous X-Ray and Flat-Top Noise Sources* (A&A) 2025

Resources on GitHub: [here](#)